

Variations on the monoidal theme

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University of Sydney Algebra Seminar

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- ▶ **Microcosm Principle** *To take the notion of monoid to its most general context requires a higher-dimensional version of monoid.*
- ▶ The principle applies again to go even higher up the stairway.
- ▶ Complicated apparently new structures such as quantum groupoids and bialgebroids become examples.
- ▶ The framework can be used to overcome obstacles caused by such things as the inability to calculate all simple objects in the centre of a semisimple monoidal category.

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- ▶ Write $\text{Mon}\mathcal{A}$ for the category of monoids in $\mathcal{A}$.
- ▶ A group G in $\mathcal{A}$ is a monoid in $\mathcal{A}$ for which the *fusion* morphism

$$G \times G \xrightarrow{1_G \times \delta} G \times G \times G \xrightarrow{\mu \times 1_G} G \times G$$

is invertible. Here $\delta : G \longrightarrow G \times G$ is the diagonal morphism.

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- ▶ More examples can be generated because: *finite product preserving functors take monoids to monoids and groups to groups.*

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Proposition

For any category $\mathcal{A}$ with finite products, evaluation at 1 determines an equivalence of categories

$$\text{FinProdPres}[\mathbb{M}, \mathcal{A}] \simeq \text{Mon} \mathcal{A} .$$

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- ▶ These isomorphisms (often called “constraints”) come as extra baggage as we ascend.

Monoidal categories

[Mac Lane 1963; Eilenberg–Kelly 1966]

- Monoidal structure on a category $\mathcal{A}$ consists of a functor

$$- \otimes - : \mathcal{A} \times \mathcal{A} \longrightarrow \mathcal{A} ,$$

natural isomorphisms $(A \otimes B) \otimes C \xrightarrow{\alpha_{A,B,C}} A \otimes (B \otimes C)$, an object I ,
and natural isomorphisms $I \otimes A \xrightarrow{\lambda_A} A \xrightarrow{\rho_A} A \otimes I$.

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- ▶ There are two axioms: the *associativity pentagon* and the *triangle for unit*.
- ▶ A monoid in **Cat** is the case where α , λ and ρ are identities.
- ▶ Every monoidal category is monoidally equivalent to a monoid in **Cat**.

Monoidal functors

- ▶ A functor $F : \mathcal{A} \longrightarrow \mathcal{X}$ between monoidal categories is *[strong] monoidal* when it is equipped with natural [iso]morphisms $\phi_{A,B} : FA \otimes FB \longrightarrow F(A \otimes B)$ and [iso]morphism $\phi_0 : I \longrightarrow FI$ satisfying three axioms like those saying the ϕ are the operations of a monoid F .

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- ▶ Monoids $M, \mu : M \otimes M \longrightarrow M, \eta : I \longrightarrow M$ can be defined in any monoidal category $\mathcal{A}$. In fact, they are monoidal functors $M : \mathbf{1} \longrightarrow \mathcal{A}$. Opmonoidal functors $C : \mathbf{1} \longrightarrow \mathcal{A}$ are *comonoids* in $\mathcal{A}$.

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- ▶ For a group G in $\mathcal{A}$, we would need a morphism $\delta : G \longrightarrow G \otimes G$ which leads to the concepts of bimonoid and Hopf monoid.

Examples of monoids in monoidal categories

Without ambiguity, we write $\mathbf{Mon}\mathcal{A}$ for the category of monoids in a monoidal category $\mathcal{A}$. If the tensor of $\mathcal{A}$ is symmetric, $\mathbf{Mon}\mathcal{A}$ is also symmetric monoidal.

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- ▶ The forgetful functor $\mathbf{Ab} \rightarrow \mathbf{Set}$ is monoidal and its left adjoint is opmonoidal.
- ▶ Monoidal functors compose and so take monoids to monoids.

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Proposition

For any monoidal category $\mathcal{A}$, evaluation at 1 determines an equivalence of categories

$$StMon[\mathbb{L}, \mathcal{A}] \simeq Mon\mathcal{A} .$$

Monoidal objects in 2-categories

- ▶ As already noted, general monoidal categories are not merely monoids in the category **Cat**. However, their structure is expressible in the *2-category* **Cat**.

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- ▶ Once this is done, we can see that such a structure can be defined in any 2-category $\mathcal{M}$ with finite products.
- ▶ We call this a *pseudomonoid* or *monoidale* in $\mathcal{M}$.

Monoidal data diagrammatically

$$\begin{array}{ccccc}
 & & \mathcal{A} \times (\mathcal{A} \times \mathcal{A}) & \xrightarrow{1 \times \otimes} & \mathcal{A} \times \mathcal{A} \\
 & \nearrow a_{\mathcal{A}, \mathcal{A}, \mathcal{A}} & & \uparrow \alpha & \searrow \otimes \\
 (\mathcal{A} \times \mathcal{A}) \times \mathcal{A} & \xrightarrow{\otimes \times 1} & \mathcal{A} \times \mathcal{A} & \xrightarrow{\otimes} & \mathcal{A}
 \end{array}$$

$$\begin{array}{ccc}
 \mathcal{A} \times \mathcal{A} & \xleftarrow{1 \times 1} & \mathbf{1} \times \mathcal{A} \\
 & \xRightarrow{\lambda} & \\
 \otimes \searrow & & \swarrow \ell \\
 & \mathcal{A} &
 \end{array}$$

$$\begin{array}{ccc}
 \mathcal{A} & \xrightarrow{r} & \mathcal{A} \times \mathbf{1} \\
 \downarrow 1 & \xRightarrow{\rho} & \downarrow 1 \times I \\
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[Joyal-St 1985]

$\text{Mon}\mathcal{A}$ is the 2-category of monoidales and monoidal morphisms in $\mathcal{A}$.

$\text{Mon}_{\text{st}}\mathcal{A}$ is the 2-category of monoidales and strong monoidal morphisms in $\mathcal{A}$.

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- ▶ Monoidales in $\text{Mon}\mathbf{Cat}$ are precisely *lax braided* monoidal categories. Monoidales in $\text{Mon}_{\text{st}}\mathbf{Cat}$ are braided monoidal categories. This is a higher Eckmann–Hilton argument and is also pertinent to homotopy theory.

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- ▶ Monoidales in $\text{Mon}_{\text{st}}\text{Mon}_{\text{st}}\mathbf{Cat}$ are *symmetric* monoidal categories.

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- ▶ Monoidales in $\text{Mon}_{\text{st}}\text{Mon}_{\text{st}}\mathbf{Cat}$ are *symmetric* monoidal categories.
- ▶ More examples can be generated because: *finite product preserving 2-functors take monoidales to monoidales*.

Invariants for 3D-manifolds: two constructions

- ▶ The Reshetikhin–Turaev [1991] invariant $\text{RT}_{\mathcal{B}}(M)$ is based on a *modular category* $\mathcal{B}$: that is, $\mathcal{B}$ is a semisimple **Vect**-enriched tortile monoidal category, with finitely many simple objects, and subject to a non-degeneracy condition. One presents the manifold M by surgery along a framed link and the link is coloured by simple objects of $\mathcal{B}$.

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- ▶ The Turaev–Viro invariant $\text{TV}_{\mathcal{C}}(M)$ is based on a *spherical category* $\mathcal{C}$: that is, a pivotal monoidal category for which the two naturally defined traces agree. Also $\mathcal{C}$ must be semisimple **Vect**-enriched with finitely many simple objects, and subject to invertibility of $\dim \mathcal{C}$. The important difference is that $\mathcal{C}$ need not be braided. One presents M by a triangulation, colours the edges by simple objects of $\mathcal{C}$ and evaluates coloured tetrahedra according to the $6j$ -symbols of $\mathcal{C}$.

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- ▶ Müger [2003] proved that such spherical $\mathcal{C}$ have their braided centres $\mathrm{Z}(\mathcal{C})$ modular.

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- ▶ The conjecture is proved by Turaev–Virelizier [2010] using ideas of Bruguières–Virelizier [2008] which stemmed from a generalization of the RT-invariant due to Lyubashenko [1995].

Invariants for 3D-manifolds: the problem

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- ▶ The problem comes in trying to apply this in the case $\mathcal{B} = \mathcal{Z}(\mathcal{C})$ for suitable spherical $\mathcal{C}$. It is not known how to obtain the simple objects in $\mathcal{Z}(\mathcal{C})$ from those in $\mathcal{C}$. What is needed is a way of calculating the cocentre of $\mathcal{B} = \mathcal{Z}(\mathcal{C})$.
- ▶ If $\mathcal{C}$ is braided and has a cocentre A then $\mathcal{Z}(\mathcal{C})$ is the category of A -modules. A substitute for this is required when $\mathcal{C}$ is not braided.

Invariants for 3D-manifolds: essential concepts

- ▶ A Hopf monoid A in braided $\mathcal{C}$ gives an opmonoidal monad $A \otimes - : \mathcal{C} \longrightarrow \mathcal{C}$. The Eilenberg-Moore algebras are the A -modules.

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- ▶ So Bruguières–Virelizier [2007] built on Moerdijk [2002] and McCrudden [2002] and defined Hopf monad on an autonomous monoidal category to be an opmonoidal monad with “antipode”. With Lack in [2010] they extended the Hopf notion to opmonoidal monads on arbitrary monoidal categories using invertibility of a fusion map. Let us look at some of these ingredients.

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- ▶ If $\mathcal{C}$ has duals and some completeness, define the *centralizer* of T by:

$$C(T)Y = \int^V (TV)^* \otimes Y \otimes V.$$

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Theorem (Bruguères–Virelizier 2008)

If $\mathcal{C}$ is autonomous, T is a Hopf monad on $\mathcal{C}$, and $C(T)$ exists, then $C(T)$ and $D(T)$ are Hopf monads on $\mathcal{C}$, and there is an equivalence of monoidal categories

$$\mathcal{C}^{D(T)} \simeq \mathcal{Z}(\mathcal{C}^T).$$

(Here $\mathcal{C}^T$ denotes the category of Eilenberg–Moore T -algebras.)

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- ▶ What is a “tensor product” on a general 2-category? Since the 2-category **2Cat** of 2-categories has products, we have the notion of monoidale in **2Cat**.
- ▶ This does actually capture what is needed for **Ab-Cat** as an example, but not for other important examples. For those we need weak 2-categories; that is, bicategories.

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 - ▶ objects denoted by U, V, W and so on,
 - ▶ hom-categories $\mathcal{W}(U, V)$ with objects denoted by arrows $f : U \longrightarrow V$ (called *1-morphisms* of $\mathcal{W}$), morphisms by double arrows $\theta : f \Longrightarrow f' : U \longrightarrow V$ (called *2-morphisms* of $\mathcal{W}$),

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$$\alpha_{k,h,g \circ f} \circ \alpha_{k \circ h,g,f} = (1_k \circ \alpha_{h,g,f}) \circ \alpha_{k,h \circ g,f} \circ (\alpha_{k,h,g} \circ 1_f)$$
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- ▶ Every bicategory is biequivalent to a 2-category. So we often ignore brackets, α , λ , and ρ .

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- ▶ The bicategory **Span** has objects sets, 1-morphisms $(u, S, v) : X \longrightarrow Y$ pairs of functions

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2-morphisms $f : (u, S, v) \Longrightarrow (s, R, t)$ functions $f : S \longrightarrow R$ such that $u = sf$ and $v = tf$, and with composition defined by pullback:

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- ▶ For any braided monoidal category $\mathcal{V}$ with reflexive equalizers preserved by tensoring with any object, there is a bicategory $\text{Comod}\mathcal{V}$. The objects are comonoids in $\mathcal{V}$. A 1-morphism $M : C \longrightarrow D$ is a left C -, right D -comodule M ; as such it has a coaction $\delta : M \longrightarrow C \otimes M \otimes D$. The 2-morphisms are comodule morphisms. Composition of 1-morphisms is defined using an equalizer. When $\mathcal{V} = \mathbf{Set}$, we have $\text{Comod}\mathcal{V} = \mathbf{Span}$.

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- ▶ Lax functors with the ϕ invertible are called *pseudofunctors*. In the one object case they are the strong monoidal functors.

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[Gordon-Power-St 1995]

- ▶ A 1-morphism $f : U \longrightarrow V$ in a bicategory $\mathcal{W}$ is an *equivalence* when there exists a 1-morphism $g : V \longrightarrow U$, and invertible 2-morphisms $g \circ f \Longrightarrow 1_U$ and $1_V \Longrightarrow f \circ g$.

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- ▶ Recently, the importance of a more general notion than monoidale has arisen in work of Kornel Szlachányi. He uses the adjective “skew”.

Skew monoidales

[Szlachanyi 2012]; [Lack-St 2012]

A *skew-monoidal* structure on an object C in a monoidal bicategory $\mathcal{W}$ consists of morphisms $p : C \otimes C \rightarrow C$, $j : I \rightarrow C$, and 2-cells

$$\begin{array}{ccccc}
 & & C \otimes (C \otimes C) & \xrightarrow{1 \otimes p} & C \otimes C \\
 & \nearrow a_{C,C,C} & & \Uparrow \alpha & \searrow p \\
 (C \otimes C) \otimes C & \xrightarrow{p \otimes 1} & C \otimes C & \xrightarrow{p} & C
 \end{array}$$

$$\begin{array}{ccc}
 C \otimes C & \xleftarrow{j \otimes 1} & I \otimes C \\
 \searrow p & \xRightarrow{\lambda} & \swarrow \ell \\
 & C &
 \end{array}$$

$$\begin{array}{ccc}
 C & \xrightarrow{r} & C \otimes I \\
 \downarrow 1 & \xRightarrow{\rho} & \downarrow 1 \otimes j \\
 C & \xleftarrow{p} & C \otimes C
 \end{array} ,$$

satisfying the following five conditions:

Pentagon (which was always really a hexagon)

$$\begin{array}{ccc}
 p \circ (p \otimes 1) \circ (1 \otimes p \otimes 1) & \xrightarrow{\alpha \circ (1 \otimes p \otimes 1)} & p \circ (1 \otimes p) \circ (1 \otimes p \otimes 1) \\
 \uparrow p \circ (\alpha \otimes 1) & & \downarrow p \circ (1 \otimes \alpha) \\
 p \circ (p \otimes 1) \circ (p \otimes 1 \otimes 1) & & p \circ (1 \otimes p) \circ (1 \otimes 1 \otimes p) \\
 \downarrow \alpha \circ (p \otimes 1 \otimes 1) & & \uparrow \alpha \circ (1 \otimes 1 \otimes p) \\
 p \circ (1 \otimes p) \circ (p \otimes 1 \otimes 1) & \xrightarrow{\cong} & p \circ (p \otimes 1) \circ (1 \otimes 1 \otimes p)
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Unit conditions

$$\begin{array}{ccc}
 p \circ (p \otimes 1) \circ (1 \otimes j \otimes 1) & \xrightarrow{\alpha \circ (1 \otimes j \otimes 1)} & p \circ (1 \otimes p) \circ (1 \otimes j \otimes 1) \\
 \uparrow p \circ (\rho \otimes 1) & & \downarrow p \circ (1 \otimes \lambda) \\
 p & \xrightarrow{1} & p
 \end{array}$$

$$\begin{array}{ccc}
 p \circ (p \otimes 1) \circ (j \otimes 1 \otimes 1) & \xrightarrow{p \circ (\lambda \otimes 1)} & p \\
 \downarrow \alpha \circ (j \otimes 1 \otimes 1) & & \uparrow \lambda \circ p \\
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Unit conditions continued

$$\begin{array}{ccc}
 p & \xrightarrow{p \circ (1 \otimes \rho)} & p \circ (1 \otimes p) \circ (1 \otimes 1 \otimes j) \\
 \rho \circ p \downarrow & & \uparrow \alpha \circ (1 \otimes 1 \otimes j) \\
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Unit conditions continued

$$\begin{array}{ccc}
 p & \xrightarrow{p \circ (1 \otimes \rho)} & p \circ (1 \otimes p) \circ (1 \otimes 1 \otimes j) \\
 \rho \circ p \downarrow & & \uparrow \alpha \circ (1 \otimes 1 \otimes j) \\
 p \circ (1 \otimes j) \circ p & \xrightarrow{\cong} & p \circ (p \otimes 1) \circ (1 \otimes 1 \otimes j)
 \end{array}$$

$$\begin{array}{ccc}
 p \circ (1 \otimes j) \circ j & \xrightarrow{\cong} & p \circ (j \otimes 1) \circ j \\
 \rho \circ j \uparrow & & \downarrow \lambda \circ j \\
 j & \xrightarrow{1} & j
 \end{array}$$

An object C of $\mathcal{W}$ equipped with a skew-monoidal structure is called a *skew monoidale* in $\mathcal{W}$.

Examples of skew monoidales

- ▶ For $\mathcal{W} = \mathbf{Cat}$ with cartesian monoidal structure, a skew monoidale is called a skew monoidal category. The definition is the same as for monoidal category except that the constraints α , λ and ρ are not required to be invertible; also, there are three more axioms (which are automatic in the non-skew case).

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- ▶ For $\mathcal{W} = \mathbf{Cat}$ with cartesian monoidal structure, a skew monoidale is called a skew monoidal category. The definition is the same as for monoidal category except that the constraints α , λ and ρ are not required to be invertible; also, there are three more axioms (which are automatic in the non-skew case).
- ▶ For $\mathcal{W} = \mathbf{Span}$ with monoidal structure given by cartesian product of sets (which is not the product in **Span**!), the skew monoidales C with unit $j = (1 \longleftarrow C \xrightarrow{1_C} C)$ are precisely categories $\mathbb{C}$. The multiplication p is forced to be a span of the form $C \times C \xleftarrow{(s,t)} A \xrightarrow{t} C$. Then C is the set of objects, A the set of morphisms, $s, t : A \longrightarrow C$ the source and target functions, and

$$p \circ (p \times 1_C) = \{x \xrightarrow{a} y \xrightarrow{b} z\} \xrightarrow{\alpha} \{x \xrightarrow{c} z \xleftarrow{b} y\} = p \circ (1_C \times p)$$

takes (a, b) to (ba, b) .

The category $\mathbb{C}$ is a groupoid if and only if α is invertible.

Bimonoid and quantum category examples

- ▶ Let A be a bimonoid in a symmetric monoidal category $\mathcal{V}$. A left skew monoidal structure on $\mathcal{V}$ with the same unit object I is defined as follows. The new tensor product is

$$X * Y = A \otimes X \otimes Y ,$$

the associativity constraint $\alpha_{XYZ} : (X * Y) * Z \longrightarrow X * (Y * Z)$ comes from the “fusion” morphism $A \otimes A \longrightarrow A \otimes A$, the left unit constraint $\lambda_X : I * X \longrightarrow X$ from the counit $\varepsilon : A \longrightarrow I$, and the right unit constraint $\rho_X : X \longrightarrow X * I$ from the unit $\eta : I \longrightarrow A$. The bimonoid is a Hopf monoid if and only if α is invertible.

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- ▶ Definitions of bialgebroids and quantum groupoids in the literature led Day-St [2004] to define quantum categories in a braided monoidal category $\mathcal{V}$. Szlachányi’s work, led Lack-St [2012] to prove that quantum categories $\mathbb{C}$ are skew monoidales in $\mathcal{W} = \text{Comod}\mathcal{V}$ with unit C as a right C -comodule using its comultiplication and counit. $\mathbb{C}$ is a quantum groupoid if and only if α is invertible.

Monoidal morphism data

Suppose C and D are skew monoidales in the monoidal bicategory $\mathcal{W}$.

A 1-morphism $f : C \longrightarrow D$ is *monoidal* when it is equipped with 2-morphisms

$$\begin{array}{ccc}
 C \otimes C & \xrightarrow{p} & C \\
 f \otimes f \downarrow & \xRightarrow{\phi} & \downarrow f \\
 D \otimes D & \xrightarrow{p} & D
 \end{array}
 \qquad
 \begin{array}{ccc}
 I & \xrightarrow{j} & C \\
 \searrow j & \xRightarrow{\phi_0} & \swarrow f \\
 & D &
 \end{array}$$

satisfying the following three axioms (reminiscent of those which would say f is a monoid with the ϕ as binary and nullary operations):

"Associativity"

$$\begin{array}{ccccc}
 (C \otimes C) \otimes C & \xrightarrow{a} & C \otimes (C \otimes C) & \xrightarrow{1 \otimes p} & C \otimes C \\
 \downarrow (f \otimes f) \otimes f & \searrow p \otimes 1 & \searrow \phi \otimes f & \xrightarrow{\alpha} & \downarrow p \\
 & & C \otimes C & \xrightarrow{p} & C \\
 & & \downarrow f \otimes f & \searrow \phi & \downarrow f \\
 (D \otimes D) \otimes D & \xrightarrow{p \otimes 1} & D \otimes D & \xrightarrow{p} & D
 \end{array}
 \quad =$$

$$\begin{array}{ccccccc}
 (C \otimes C) \otimes C & \xrightarrow{a} & C \otimes (C \otimes C) & \xrightarrow{1 \otimes p} & C \otimes C & & \\
 \downarrow (f \otimes f) \otimes f & & \downarrow f \otimes (f \otimes f) & \searrow f \otimes \phi & \downarrow f \otimes f & \searrow p & \\
 & & D \otimes (D \otimes D) & \xrightarrow{1 \otimes p} & D \otimes D & \xrightarrow{\phi} & C \\
 \nearrow a & & \cong & \searrow \alpha & \nearrow p & \searrow f & \\
 (D \otimes D) \otimes D & \xrightarrow{p \otimes 1} & D \otimes D & \xrightarrow{p} & D & &
 \end{array}$$

“Left unit condition”

$$\begin{array}{ccccc}
 I \otimes C & \xrightarrow{\ell} & C & & \\
 \downarrow 1 \otimes f & \searrow j \otimes 1 & \nearrow p & & \downarrow f \\
 & C \otimes C & & & \\
 & \downarrow f \otimes f & & & \\
 I \otimes D & \xrightarrow{\ell} & D & & \\
 \downarrow 1 \otimes f & \searrow j \otimes f & \nearrow p & & \downarrow f \\
 & D \otimes D & & & \\
 & \downarrow p & & & \\
 & D & & &
 \end{array}$$

$\phi_0 \otimes f \Rightarrow$ (arrow from $I \otimes C$ to $C \otimes C$)
 $\lambda \uparrow$ (arrow from $C \otimes C$ to C)
 $\phi \Rightarrow$ (arrow from $C \otimes C$ to $D \otimes D$)
 $\lambda \otimes f \Rightarrow$ (arrow from $I \otimes D$ to $D \otimes D$)

=

$$\begin{array}{ccc}
 I \otimes C & \xrightarrow{\ell} & C \\
 1 \otimes f \downarrow & \cong & \downarrow f \\
 I \otimes D & \xrightarrow{\ell} & D
 \end{array}$$

“Right unit condition”

$$\begin{array}{ccc}
 D & \xleftarrow{f} & C \\
 \downarrow 1 & \searrow r & \downarrow r \\
 & D \otimes I & \xleftarrow{f \otimes 1} C \otimes I \\
 & \downarrow 1 \otimes j & \downarrow 1 \otimes j \\
 & D \otimes D & \xleftarrow{f \otimes f} C \otimes C \\
 & \swarrow p & \downarrow p \\
 D & \xleftarrow{f} & C
 \end{array}
 \begin{array}{c}
 \cong \\
 \xRightarrow{\rho} \\
 \xRightarrow{f \otimes \phi_0} \\
 \xRightarrow{\phi}
 \end{array}
 \begin{array}{ccc}
 C & \xrightarrow{r} & C \otimes I \\
 \downarrow f & \xRightarrow{f\rho} & \downarrow 1 \otimes j \\
 & C \otimes C & \\
 & \downarrow p & \\
 D & \xleftarrow{f} & C
 \end{array}$$

Skew duoidales

Lack-St [preprint 2012]

- ▶ If $\mathcal{W}$ is a symmetric monoidal bicategory, we obtain a symmetric monoidal bicategory $\text{SkMon}\mathcal{W}$ whose objects are skew monoidales in $\mathcal{W}$, whose morphisms are monoidal morphisms, and whose 2-morphisms are the obvious thing. So the process can be repeated.

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- ▶ An object of $\text{SkMonSkMon}\mathcal{W}$ is called a *skew duoidale*.
- ▶ An Eckmann-Hilton-style argument for skew duoidales is used to show that,
in a cartesian bicategory [Carboni-Kelly-Walters-Wood 2010], *quantum categories are exactly monads*.